Do brains synchronize more to biological motion?
An inter-subject correlation (ISC) analysis of EEG whole-body movement data

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Introduction (revised from last submission)

People are highly sensitive to biological motion, the dynamic visual patterns associated with the movements of living entities. For example, we can recognize human movement from just a few moving dots placed on the major joints of the body (Johansson, 1973), known as point-light displays (PLDs). In cognitive neuroscience, PLDs have been used to measure which areas of the brain are active during biological motion by comparing coherent, meaningful PLDs with "scrambled" versions in which the dot starting locations have been randomized. This comparison ensures that researchers isolate neural responses to the global biological motion stimulus, rather than low-level visual motion cues. For each individual, responses are averaged within each channel and time window, and these averages are then compared between conditions using inferential statistics (e.g., *t* test, ANOVA). Functional magnetic resonance imaging (fMRI) research using this approach has identified responses to biological motion in visual and sensorimotor regions (Grosbras et al., 2012), with signals in these regions emerging from roughly 200-360 ms after stimulus onset in electroencephalography (EEG) and magnetoencephalography (MEG) (Chang et al., 2021; Hirai et al., 2003; Jokisch et al., 2005).

Although this approach has been successful in capturing the average brain response to biological motion, other research using naturalistic stimuli suggests that dynamic, socially relevant stimuli evoke neural responses with a shared temporal structure across viewers (Nastase et al., 2019). In fMRI studies using audiovisual movie clips, the shared pattern of neural activity across individuals has been linked to the effectiveness of interpersonal communication (Stephens et al., 2010). More recently, this shared response has also been measured with EEG, where it can provide information about attentional engagement (Dmochowski et al., 2012) and audience preferences for advertisements and television scenes (Dmochowski et al., 2014). These approaches use intersubject correlation (ISC), a measure of

how similar the activity in a given brain region is across people over time while they experience the same naturalistic stimulus (Hasson et al., 2004, 2010). Participants receive the same time-locked stimulus, which allows the degree of shared response to be compared between experimental conditions. Then, the response time course of each brain region or electrode in one subject is correlated with the corresponding time courses of the other subjects, rather than correlating time series across different electrodes within a single subject (Nastase et al., 2019; Simony et al., 2016). Therefore, a low ISC signals an inconsistent pattern of response across people, rather than that the brain is not responding.

Because biological motion carries dynamic social information, it is a good candidate for ISC analysis. However, it remains unclear whether PLDs, which reduce biological motion to a few moving dots, elicit the same kind of shared neural response as more complex naturalistic stimuli. This project will try to address whether such minimal displays are sufficient to drive shared neural responses across viewers by measuring ISC in EEG data during observation of PLDs of whole-body human movements. EEG was recorded while participants viewed PLDs of coherent, whole-body actions and scrambled versions of the same stimuli in which the starting locations of the dots were randomized. Scrambling keeps each dot's movement the same but breaks up the shape of the body, so comparing the two conditions shows the effect of seeing a whole human figure.

ISC, which quantifies how closely each participant's neural activity tracks that of the other viewers, will be computed in MATLAB using Dmochowski's Reliable Components Analysis (RCA) toolbox (Dmochowski et al., 2012, 2015) and compared between the two conditions. EEG electrodes record a mixture of many neural sources and noise, and RCA combines the electrodes into a few components that make the response as consistent as possible across participants. RCA concentrates the stimulus-driven signal, reduces the

number of statistical comparisons, and gives each component a scalp topography showing where the shared response is strongest.

The hypothesis is that ISC will be higher for coherent motion than for scrambled motion. Higher ISC would not mean that coherent motion simply produces more brain activity, but that more of the neural response is driven by the stimulus in the same way across viewers, suggesting that the human form itself is processed consistently across people.

Methods

Participants

The data comes from a published study (Harris et al., 2025). The study included 139 participants, aged 18-30, recruited from the five Claremont Colleges. Thirty participants were excluded due to technical problems with the EEG recording (n = 10), software or piloting issues (n = 10), excessive eye-blink or motor artifacts (n = 5), or chance-level performance on the one-back task ($d' \leq 0$) in at least one block (n = 5). The final sample includes 109 participants (60 female, 48 male and 1 nonbinary out of which 103 were right-handed).

Stimuli

Stimuli were 3s point-light displays (PLDs) of whole-body human movements taken by male and female actors. The emotional movements consisted of 18 videos: six performances each of three emotions (happiness, anger, and sadness). The neutral movements consisted of six videos: two performances each of three neutral actions (marching, hopping, and toe touches). Each of the 24 coherent PLDs was also presented in a scrambled version where the starting location of each dot was randomized. This keeps the low-level motion cues but makes the movements no longer look like a human action.

Procedure

Participants completed the observation task together with a bimanual finger-tapping task. For 55 participants, the observation task came between two sets of finger-tapping blocks, while the other 54 completed it before any finger tapping. Coherent and scrambled PLDs were presented in two separate blocks, with block order counterbalanced across participants. Each block contained 108 trials with three presentations of each of the 18 emotional videos (54 trials) and nine presentations of each of the six neutral videos (54 trials). Each video was followed by a 2s pause with the final frame remaining on screen. To keep attention equal across conditions, participants performed a one-back task, pressing a button during the inter-trial interval whenever the same video appeared twice in a row. The trial order was pseudo-randomized so that no video appeared more than twice in a row. EEG was recorded while participants watched the videos using a high-density 128-channel BioSemi ActiveTwo system, with additional electrodes on both mastoids for reference. Data was recorded continuously at 512 Hz. In MATLAB, EEG data will then be split up into trials using the event triggers, separated by condition and sorted by stimulus to align participants for ISC.

ISC Analysis

Inter-subject correlation (ISC) measures how similarly different people's brains respond to the same time-locked stimulus (Hasson et al., 2004). Following Nastase et al. (2019), the signal at each electrode can be split into three parts: a stimulus-driven response shared by all viewers; an idiosyncratic response to the stimulus that differs between people in timing or strength, for example because of individual interpretations or memories; activity unrelated to the stimulus, such as mind-wandering or physiological and noise.

Only the shared part is what is common to everyone. When correlating the people's time courses with each other, the other two parts cancel out. ISC therefore tells us how much of the total variance in the signal is shared between viewers. The larger the shared part relative to the idiosyncratic response and the noise, the higher the ISC. An ISC of 1 means that the stimulus is processed by the views in exactly the same way. Values near 0 mean that the responses are idiosyncratic or carry little information about the stimulus.

EGG from 250-600ms after video onset was analysed using Dmochowski's RCA toolbox in MATLAB (Dmochowski et al., 2012, 2015). Using the time frame we are using, the window is 179 samples at 512 Hz. RCA combines the 128 electrodes into weighted sums called components. The weights are chosen so that the part of the signal that repeats on every trial is as large as possible compared with the noise that changes randomly from trial to trial. The components are ranked by how their responses repeat across trials. The three reliable components (RC1, RC2 and RC3) were kept. For each one a scalp map shows how strongly each electrode picks up that component, which shows where on the scalp its activity is the strongest.

For each participant, condition and component, the time course was averaged across trials. ISC was then computed as the Pearson correlation between this average and the mean response of all other participants, giving us one ISC value per participant, condition and component (leave-one-out). A higher value means that a person's response was more similar to the group's.

Correlations can only range from -1 to 1 and therefore all ISC values were Fisher z-transformed before testing, making them normally distributed. Then, for each component, the coherent and scrambled conditions were compared with … [to be continued: t-test or Wilcoxon signed-rank test ?]